MANAGEMENT OPERATING REVIEW - RECORD 024

Helioscope Robotics, Inc. - diligence file reference OPERATING_REVIEWS-024

Quarterly Business Review for Q1 2026: purpose and attendance

Meeting discipline held: sections that had no decision to make were noted as read and skipped in the room. Tobias Nkemdirim, VP Engineering, presented the operations section. Hollis Bramwell attended as Director of Supply Chain. The cadence changed to six-weekly after this. This was the final monthly of the cycle. The Quarterly Business Review convened for Q1 2026. Notes were circulated on 2026-04-23. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Notes were captured live and circulated the same day, which ended a habit of arguing about what was decided. Pre-reads went out ahead of the 2026-04-23 session. The Quarterly Business Review convened for Q1 2026. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Hollis Bramwell attended as Director of Supply Chain. Notes were circulated on 2026-04-23. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Guests from the deployment crews attended by video and contributed the only new information in the operations section. Notes were circulated on 2026-04-23. Tobias Nkemdirim, VP Engineering, presented the operations section. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Pre-reads went out ahead of the 2026-04-23 session. Hollis Bramwell attended as Director of Supply Chain. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Reading the packet aloud was abandoned two sessions ago and nobody has asked for it back. One function submitted its pre-read late. The Quarterly Business Review convened for Q1 2026. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Tobias Nkemdirim, VP Engineering, presented the operations section. The next session is set for 2026-05-14. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Deployment operations

Parallel deployment at two sites in the same region worked for the first time, which changes what the queue can absorb. Queue depth reached 37 sites. Contract to go-live averaged 37 days. The cadence changed to six-weekly after this. This was the final monthly of the cycle. The queue carries 37 sites, 34 a period ago. Salinas Valley was held at the readiness gate. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

The queue holds three sites with no confirmed dock date from the customer, and the review declined to schedule crews against hope. Go-live cycle: 37 days, from 38. Waukegan and Ashland Park both cleared acceptance testing. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Contract to go-live averaged 37 days. Hollis Bramwell sequenced the queue by readiness. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Days from contract to go-live improved, but the improvement came from the front of the process rather than from anything the installation crews changed. Hollis Bramwell sequenced the queue by readiness. Salinas Valley was held at the readiness gate. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Waukegan and Ashland Park both cleared acceptance testing. Queue depth reached 37 sites. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

The deployment playbook was rewritten mid-period after a crew found that the network prerequisites section was three revisions out of date. Waukegan and Ashland Park both cleared acceptance testing. Queue depth reached 37 sites. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Hollis Bramwell sequenced the queue by readiness. Salinas Valley was held at the readiness gate. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Crew travel days per site came down, largely because two regional hubs now hold enough tooling to avoid shipping kit ahead of each job. Hollis Bramwell sequenced the queue by readiness. Waukegan and Ashland Park both cleared acceptance testing. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Contract to go-live averaged 37 days. Salinas Valley was held at the readiness gate. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

Acceptance testing found more integration defects than mechanical ones, and the defects clustered in the warehouse-management adapter. Hollis Bramwell sequenced the queue by readiness. Salinas Valley was held at the readiness gate. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Go-live cycle: 37 days, from 38. Contract to go-live averaged 37 days. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Customer escalations live at the Q1 2026 review

Account	Issue	Opened	Severity	State	Next step
Northstar Cold Chain	interface timeouts to the host	2026-04-11	high	monitoring	parts shipped
Havering Logistics	sensor head replacement scheduling	2026-04-02	low	escalated to product	parts shipped
Quillon Home Supply	operator training gap	2026-03-24	high	open	definition agreed
Tenmile Freight	uptime reporting dispute	2026-03-15	low	mitigated	parts shipped
Bexley Pharmaceutical Wholesale	docking misalignment after release	2026-03-06	high	closed	parts shipped
Marrowfield Grocers	aisle congestion at peak	2026-02-25	low	monitoring	parameter set applied
Pinebank Distribution	charger fault on two units	2026-02-16	high	escalated to product	site visit booked
Alder & Vance Retail	interface timeouts to the host	2026-02-07	low	open	parts shipped
Northstar Cold Chain - second site	sensor head replacement scheduling	2026-01-29	high	mitigated	joint review scheduled
Havering Logistics - second site	operator training gap	2026-01-20	low	closed	no further action
Quillon Home Supply - second site	uptime reporting dispute	2026-01-11	high	monitoring	parts shipped
Tenmile Freight - second site	docking misalignment after release	2026-01-02	low	escalated to product	parameter set applied
Bexley Pharmaceutical Wholesale - second site	aisle congestion at peak	2025-12-24	high	open	no further action
Marrowfield Grocers - second site	charger fault on two units	2025-12-15	low	mitigated	no further action
Pinebank Distribution - second site	interface timeouts to the host	2025-12-06	high	closed	definition agreed

Deployment resourcing

Handover from crew to service is still informal, and two accounts have complained about who to call after go-live. Owen Castellanos covered two regions this period. Utilisation was 85 percent against 83 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Installer utilisation ran at 85 percent. Travel days per site

fell again. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Contractor coverage buys schedule protection at a cost per site that the review agreed to carry for a bounded number of jobs. Travel days per site fell again. Utilisation was 85 percent against 83 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Grand Forks needed a second crew visit. Owen Castellanos covered two regions this period. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Regional crew hubs changed the utilisation picture more than any hiring decision made in the same period. Utilisation was 85 percent against 83 percent. Travel days per site fell again. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Crew leads remain the gating resource. Grand Forks needed a second crew visit. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Weekend and holiday work is filling gaps that the capacity model does not show, which makes the model look better than reality. Travel days per site fell again. Crew leads remain the gating resource. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Installer utilisation ran at 85 percent. Grand Forks needed a second crew visit. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Crew composition changed this period, with more integration skill and less mechanical assembly skill in each team. Grand Forks needed a second crew visit. Installer utilisation ran at 85 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Utilisation was 85 percent against 83 percent. Travel days per site fell again. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

Scheduling crews more than six weeks out has proved futile, because customer readiness dates move inside that window. Travel days per site fell again. Utilisation was 85 percent against 83 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Owen Castellanos covered two regions this period. Installer utilisation ran at 85 percent. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Spares depot performance by part class, Q1 2026

Depot	Part class	Part lines stocked	Fill rate %	Backorders	Repair backlog units
Reno depot	mechanical	176	100	10	19
Columbus depot	mechanical	143	96	5	1
Dallas depot	mechanical	222	100	12	11
Atlanta depot	mechanical	217	94	1	0
Toronto depot	mechanical	198	99	9	13
Vehicle stock	mechanical	193	92	4	3
Reno depot	electrical and sensing	244	100	11	17
Columbus depot	electrical and sensing	239	95	2	7
Dallas depot	electrical and sensing	206	100	7	21
Atlanta depot	electrical and sensing	285	97	0	0
Toronto depot	electrical and sensing	238	100	8	15
Vehicle stock	electrical and sensing	261	93	3	5

Manufacturing output

The build plan was cut mid-period rather than allowed to fail quietly, and the cut was communicated to deployment the same week. Attainment gap: 2 units short of plan. Output reached 110 units, from 104. The cadence changed to six-weekly after this. This was the final monthly of the cycle. The Beacon WMS adapter drove most of the changeover time. Plan was 112 units; the line delivered 110. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Second-shift ramp is going more slowly than planned because training the new operators takes line time from the first shift. Attainment gap: 2 units short of plan. Emil Farkas owns the line rebalance. The cadence changed to six-weekly

after this. This was the final monthly of the cycle. Plan was 112 units; the line delivered 110. Output reached 110 units, from 104. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Test-cell capacity will bind before assembly capacity does, and the capital request to fix that has already been deferred once. Output reached 110 units, from 104. Plan was 112 units; the line delivered 110. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Emil Farkas owns the line rebalance. The Beacon WMS adapter drove most of the changeover time. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Work instructions were re-photographed for the revised sensor head, ending a period where operators worked from a printout with a pen annotation. Emil Farkas owns the line rebalance. Overtime carried the closing two weeks. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Output reached 110 units, from 104. Plan was 112 units; the line delivered 110. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Rebalancing the two assembly cells lifted output without adding people, mostly by moving the harness dress-up operation upstream. Emil Farkas owns the line rebalance. Plan was 112 units; the line delivered 110. The cadence changed to six-weekly after this. This was the final monthly of the cycle. The Beacon WMS adapter drove most of the changeover time. Work in process ended at 46 units. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

Cycle time on the pallet mover remains longer than the tote handler by more than the work content justifies. Work in process ended at 46 units. Attainment gap: 2 units short of plan. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Emil Farkas owns the line rebalance. Plan was 112 units; the line delivered 110. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Quality performance

Inspection sampling reverted to the standard plan with a documented trigger, so the next excursion will be caught earlier. Lucia Mbeki-Hart signed the corrective action. Yield was 97 percent against 96 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. First-pass yield finished at 97 percent. Scrap ran 13 thousand, from 16 thousand. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Quality engineering asked for one more headcount and got a re-prioritisation instead, which the review acknowledged as a real trade. Tessellate Controls changed a process without notice. First-pass yield finished at 97 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Yield was 97 percent against 96 percent. Scrap totalled 13 thousand dollars. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Process capability studies were completed on the two operations that had never had one, and one of them failed. Yield was 97 percent against 96 percent. Lucia Mbeki-Hart signed the corrective action. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Scrap totalled 13 thousand dollars. First-pass yield finished at 97 percent. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Containment held: nothing built in the affected window left the plant without a hundred percent inspection record attached. Scrap ran 13 thousand, from 16 thousand. Lucia Mbeki-Hart signed the corrective action. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Rework hours concentrated in the sensor cell. Yield was 97 percent against 96 percent. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Escapes to the field are the measure that matters, and this period produced a number that the service team could absorb. Lucia Mbeki-Hart signed the corrective action. Tessellate Controls changed a process without notice. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Rework hours concentrated in the sensor cell. Scrap ran 13 thousand, from 16 thousand. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

A supplier process change arrived without notification and produced a yield loss that took five weeks to attribute correctly. First-pass yield finished at 97 percent. Tessellate Controls changed a process without notice. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Rework hours concentrated in the sensor cell. Scrap ran 13 thousand, from 16 thousand. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Warranty claims by build vintage and line, Q1 2026

Build vintage	Product line	Fleet units	Claims in period	Warranty cost per unit USD	Trend
2023 build	HS-200 tote handler	43	1	202	first period measured
2024 first half build	HS-200 tote handler	84	13	312	improving
2024 second half build	HS-200 tote handler	98	9	290	flat
2025 first half build	HS-200 tote handler	127	20	246	worsening
2025 second half build	HS-200 tote handler	156	12	224	improving sharply
2026 build	HS-200 tote handler	173	22	290	first period measured
2023 build	HS-400 pallet mover	58	4	268	improving
2024 first half build	HS-400 pallet mover	81	14	202	flat
2024 second half build	HS-400 pallet mover	101	7	312	worsening
2025 first half build	HS-400 pallet mover	124	14	224	improving sharply
2025 second half build	HS-400 pallet mover	159	11	246	first period measured
2026 build	HS-400 pallet mover	182	22	268	improving

Inbound supply position

Buyers are chasing acknowledgements on a fixed cadence now, and the cadence has already caught two silent date pushes. Brightwater Motion missed two consecutive commit dates. On-time delivery was 96 percent, from 95 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Expedite activity rose with the build plan of 112. Calder Battery Works improved after a site visit. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Freight mode is a planning outcome rather than a logistics choice, and the air spend proves it. Tobias Nkemdirim issued the first supplier scorecards. Expedite activity rose with the build plan of 112. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Supplier on-time delivery came in at 96 percent. Calder Battery Works improved after a site visit. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Expedite spend is the honest measure of how well planning is working, and it went the wrong way this period. Expedite activity rose with the build plan of 112. Brightwater Motion missed two consecutive commit dates. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Calder Battery Works improved after a site visit. Supplier on-time delivery came in at 96 percent. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

One casting supplier has become a constraint on the pallet mover, and the tooling is theirs rather than ours. Supplier on-time delivery came in at 96 percent. On-time delivery was 96 percent, from 95 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Coverage weeks on long-lead parts were extended. Expedite activity rose with the build plan of 112. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar

assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

A battery cell allocation notice arrived with thirty days of warning, and the planning team had inventory to absorb it. Supplier on-time delivery came in at 96 percent. On-time delivery was 96 percent, from 95 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Calder Battery Works improved after a site visit. Brightwater Motion missed two consecutive commit dates. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

Inbound freight mode shifted toward air on two part numbers, which protected the build plan and shows up in cost per unit. Coverage weeks on long-lead parts were extended. Tobias Nkemdirim issued the first supplier scorecards. The cadence changed to six-weekly after this. This was the final monthly of the cycle. On-time delivery was 96 percent, from 95 percent. Calder Battery Works improved after a site visit. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Constraint tracking since the last session

Constraint under review: cash and collections, at the discipline held stage as of this session. Runway improved by four months against the plan struck a year earlier, driven roughly two-thirds by collections and one-third by the deferred test-cell capex. Go-live cycle 37 days at Ashland Park. Slipped deals 14, including Bexley Pharmaceutical Wholesale.

Freight mode is a planning outcome rather than a logistics choice, and the air spend proves it. Coverage weeks on long-lead parts were extended. Calder Battery Works improved after a site visit. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Tobias Nkemdirim issued the first supplier scorecards. Supplier on-time delivery came in at 96 percent. Crew Bravo closed Grand Forks without a second visit. Yield 97 percent, scrap 13 thousand in Harness and electrical. Depot fill 98 percent at Reno depot. Requisitions open 20, Hollis Bramwell holding the list.

Expedite spend is the honest measure of how well planning is working, and it went the wrong way this period. Brightwater Motion missed two consecutive commit dates. On-time delivery was 96 percent, from 95 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Tobias Nkemdirim issued the first supplier scorecards. Calder Battery Works improved after a site visit. Capital spend 890 thousand this period. Salinas Valley carries 37 of the queued fleet. Uptime 98.6 percent, interventions every 433 hours. The HS-400 pallet mover took most of the test cell time.

Headcount plan against actual by function and location, Q1 2026

Function	Location	Plan	Actual	Gap	Open requisitions
Engineering	headquarters and plant	42	33	17	4
Manufacturing	headquarters and plant	33	23	16	0
Deployment and field service	headquarters and plant	21	12	15	4
Commercial	headquarters and plant	12	8	7	3
Customer success	headquarters and plant	8	7	3	5
Supply chain	headquarters and plant	7	5	3	2
Finance and administration	headquarters and plant	3	52	0	1
People	headquarters and plant	55	40	26	1
Engineering	field and regional	24	17	8	3
Manufacturing	field and regional	18	12	8	0
Deployment and field service	field and regional	12	6	7	3
Commercial	field and regional	7	4	3	2
Customer success	field and regional	5	3	1	4

Function	Location	Plan	Actual	Gap	Open requisitions
Supply chain	field and regional	3	2	1	1
Finance and administration	field and regional	1	27	0	0
People	field and regional	31	20	13	0

Stock coverage and ageing

Receiving is holding returned parts that could be reworked, and nobody owns the decision to rework them. Work in process closed at 46 units. Work in process moved from 41 to 46 units. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Buffer stock on the drive unit was raised deliberately. Kit shortages were concentrated in low-value parts. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Inventory is growing faster than build volume, and the growth is in long-lead components rather than in finished goods. Floor space in staging is now the binding limit. Work in process closed at 46 units. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Kit shortages were concentrated in low-value parts. Buffer stock on the drive unit was raised deliberately. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Finished-goods staging is now constrained by floor space, which is an odd problem to have while missing the build plan. Excess exposure sits in the superseded sensor head. Floor space in staging is now the binding limit. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Work in process moved from 41 to 46 units. Hollis Bramwell owns the disposition decision. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Cycle counting accuracy improved after the bin relabelling, and the count variance is now small enough to trust the system quantity. Work in process moved from 41 to 46 units. Work in process closed at 46 units. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Hollis Bramwell owns the disposition decision. Excess exposure sits in the superseded sensor head. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Kit shortages are concentrated in low-value parts, which is the most annoying possible failure mode. Work in process closed at 46 units. Floor space in staging is now the binding limit. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Hollis Bramwell owns the disposition decision. Buffer stock on the drive unit was raised deliberately. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

Coverage weeks by commodity is the only inventory number this meeting has ever acted on, so the others were dropped from the packet. Floor space in staging is now the binding limit. Excess exposure sits in the superseded sensor head. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Kit shortages were concentrated in low-value parts. Buffer stock on the drive unit was raised deliberately. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Service performance across the fleet

Software releases are now staged across sites rather than pushed at once, after a release regressed docking behaviour at two accounts. Owen Castellanos leads the joint remediation plan. Havering Logistics measures uptime on a stricter basis. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Fleet uptime averaged 98.6 percent. Waukegan remains the weakest site on interventions. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Remote diagnostics closed more incidents without a visit than in any prior period, which is where the cost curve bends. Waukegan remains the weakest site on interventions. Mean time between interventions was 433 hours. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Havering Logistics measures uptime on a stricter basis. Interventions spaced 433 hours apart on the fleet median. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Reliability engineering split the top failure modes by build vintage and found that one of them is entirely a legacy population problem. Mean time between interventions was 433 hours. Owen Castellanos leads the joint remediation plan. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Interventions spaced 433 hours apart on the fleet median. Havering Logistics measures uptime on a stricter basis. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Fleet uptime across all live sites improved, and the improvement is broad rather than driven by one recovering account. Owen Castellanos leads the joint remediation plan. Uptime reached 98.6 percent, from 98.5 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Havering Logistics measures uptime on a stricter basis. Mean time between interventions was 433 hours. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Site-level variation in reliability is wider than the fleet average suggests, and the worst site drags the contractual conversation. Waukegan remains the weakest site on interventions. Mean time between interventions was 433 hours. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Fleet uptime averaged 98.6 percent. Uptime reached 98.6 percent, from 98.5 percent. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

The remediation plan with the worst-performing account is joint, with obligations written on both sides and a fixed measurement window. Mean time between interventions was 433 hours. Owen Castellanos leads the joint remediation plan. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Uptime reached 98.6 percent, from 98.5 percent. Waukegan remains the weakest site on interventions. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Requisition ageing and decisions, Q1 2026

Requisition	Role	Days open	Stage	Recruiting owner	Decision this review
REQ-2440	Field engineer	24	offer out	hiring manager sourced	continue
REQ-2442	Crew lead	57	offer declined	external agency	raise band
REQ-2444	Controls engineer	38	on hold	external agency	widen radius
REQ-2446	Manufacturing technician	67	closed without hire	internal recruiting	pause
REQ-2448	Buyer and planner	48	sourcing	external agency	close and re-scope
REQ-2450	Solutions architect	77	phone screen	Renata Voss	convert contractor
REQ-2452	Service dispatcher	28	on-site loop	Renata Voss	continue
REQ-2454	Reliability engineer	61	offer out	Renata Voss	raise band
REQ-2456	Field engineer (second opening)	42	offer declined	hiring manager sourced	widen radius
REQ-2458	Crew lead (second opening)	71	on hold	hiring manager sourced	pause
REQ-2460	Controls engineer (second opening)	36	closed without hire	external agency	close and re-scope
REQ-2462	Manufacturing technician (second opening)	57	sourcing	Renata Voss	convert contractor
REQ-2464	Buyer and planner (second opening)	24	phone screen	Renata Voss	continue
REQ-2466	Solutions architect (second opening)	57	on-site loop	external agency	raise band
REQ-2468	Service dispatcher (second opening)	38	offer out	Renata Voss	widen radius

Requisition	Role	Days open	Stage	Recruiting owner	Decision this review
REQ-2470	Reliability engineer (second opening)	67	offer declined	internal recruiting	pause
REQ-2472	Field engineer (backfill)	48	on hold	internal recruiting	close and re-scope

Depot fill and warranty

Service parts pricing has never been reviewed, and two parts are being sold below their landed cost. Fill rate ran 98 percent against 96 percent. Return material flow remains slow. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Ashland Park took the planned sensor head replacements. Depot fill rate was 98 percent. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Obsolescence on the first-generation controller will force a spares buy decision before the end of next year. Warranty cost per fleet unit is falling. Return material flow remains slow. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Emil Farkas approved the critical spares increase. Vehicle stock is masking the true fill rate. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

First-generation controllers are heading for obsolescence, and the spares buy has to be decided before the line stops making them. Warranty cost per fleet unit is falling. Fill rate ran 98 percent against 96 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Return material flow remains slow. Emil Farkas approved the critical spares increase. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Warranty accrual assumptions were set before the yield excursion and have not been revisited since. Fill rate ran 98 percent against 96 percent. Depot fill rate was 98 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Warranty cost per fleet unit is falling. Ashland Park took the planned sensor head replacements. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Return material flow is slow, and the returned parts that could be reworked are sitting in a corner of receiving. Ashland Park took the planned sensor head replacements. Depot fill rate was 98 percent. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Vehicle stock is masking the true fill rate. Return material flow remains slow. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

Critical spares coverage for the drive unit was raised deliberately while the supply constraint was live. Ashland Park took the planned sensor head replacements. Return material flow remains slow. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Fill rate ran 98 percent against 96 percent. Vehicle stock is masking the true fill rate. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Leavers and backfill position, Q1 2026

Function	Leavers in period	Annualised attrition %	Primary stated reason	Backfill position
Engineering	3	16	career change outside the industry	contractor covering
Manufacturing	1	7	manager relationship	deferred to next quarter
Deployment and field service	3	12	retirement	requisition open
Commercial	0	5	performance	backfilled internally
Customer success	2	13	travel load	not backfilling
Supply chain	0	8	compensation	contractor covering
Finance and administration	2	15	relocation	deferred to next quarter
People	1	10	career change outside the industry	requisition open
Engineering (contract)	3	17	manager relationship	backfilled internally
Manufacturing (contract)	1	6	retirement	not backfilling
Deployment and field service (contract)	4	14	performance	contractor covering

Function	Leavers in period	Annualised attrition %	Primary stated reason	Backfill position
Commercial (contract)	0	9	travel load	deferred to next quarter
Customer success (contract)	3	16	compensation	requisition open

Hiring and attrition

Contractor-to-employee conversion is the cheapest source of certified crew, and the process for it does not exist yet. Median time to fill was 48 days. Time to fill improved from 54 to 48 days. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Requisitions fell from 24 to 20. Open requisitions number 20. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Headcount against plan closed part of its gap, and the remaining gap is concentrated in the two functions that gate deployment capacity. Annualised attrition sits at 11 percent. Time to fill improved from 54 to 48 days. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Requisitions fell from 24 to 20. Headcount stood at 278 against a plan of 290. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Open requisitions were re-ranked against the operating plan, and four were closed rather than left to age quietly. The headcount gap is 12 roles. Headcount stood at 278 against a plan of 290. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Time to fill improved from 54 to 48 days. Annualised attrition sits at 11 percent. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Manager span of control has grown past what the review considers sustainable in two functions. Open requisitions number 20. Headcount stood at 278 against a plan of 290. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Annualised attrition sits at 11 percent. The headcount gap is 12 roles. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

A hiring pause was applied selectively, protecting customer-facing roles and holding back internal ones. Median time to fill was 48 days. Headcount stood at 278 against a plan of 290. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Annualised attrition sits at 11 percent. Requisitions fell from 24 to 20. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

Engagement survey results split by tenure rather than by function, with the newest cohort least positive about clarity of direction. Open requisitions number 20. The headcount gap is 12 roles. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Headcount stood at 278 against a plan of 290. Median time to fill was 48 days. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Invoicing and days outstanding

Aged balances were read out account by account, which took ten minutes and closed two of them the same week. Two receivables aged past ninety days. Receipts of 10240 thousand outpaced invoicing. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Outstanding days improved from 42 to 39. Lucia Mbeki-Hart moved collections calls to the account teams. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Days sales outstanding is the metric under active management, and the improvement came from the invoicing process rather than from customer behaviour. Bexley Pharmaceutical Wholesale requires a separate submission path. Lucia Mbeki-Hart moved collections calls to the account teams. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Collections totalled 10240 thousand dollars. Days sales outstanding closed at 39 days. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Two large receivables aged past ninety days, and both trace to a purchase order number mismatch rather than to unwillingness to pay. Two receivables aged past ninety days. Receipts of 10240 thousand outpaced invoicing. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Days sales outstanding closed at 39 days. Collections totalled 10240 thousand dollars. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Payment terms granted in commercial negotiation were never visible to the finance team until invoicing, and that has been fixed. Collections totalled 10240 thousand dollars. Outstanding days improved from 42 to 39. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Lucia Mbeki-Hart moved collections calls to the account teams. Two receivables aged past ninety days. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Bad debt remains immaterial, and the review noted that this is a function of customer quality rather than of credit discipline. Receipts of 10240 thousand outpaced invoicing. Two receivables aged past ninety days. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Days sales outstanding closed at 39 days. Bexley Pharmaceutical Wholesale requires a separate submission path. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Headcount 278 of 290, Tobias Nkemdirim reporting. Burn 10860 thousand, runway 19 months.

Getting the acceptance document signed on the day of go-live is worth more to cash than any collections call. Receipts of 10240 thousand outpaced invoicing. Outstanding days improved from 42 to 39. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Bexley Pharmaceutical Wholesale requires a separate submission path. Lucia Mbeki-Hart moved collections calls to the account teams. On-time delivery 96 percent, Tessellate Controls the outlier. Uptime 98.6 percent, interventions every 433 hours. Capital spend 890 thousand this period. Harness and electrical ran 97 percent first pass.

Burn by category, Q1 2026

Category	Period USD thousands	Prior period USD thousands	Movement %	Commentary
Engineering payroll	1635	253	546.2	headcount ramp
Manufacturing payroll and overheads	977	246	297.2	spares buy for the drive unit
Deployment labour, employees	868	176	393.2	second hub fit-out
Deployment labour, contract crews	760	211	260.2	overtime in the closing weeks
Commercial payroll	434	140	210.0	test cell deferred
Marketing and events	651	176	269.9	headcount ramp
Field service payroll	543	140	287.9	test cell deferred
Spares and consumables	543	105	417.1	second hub fit-out
Facilities and rent	434	70	520.0	spares buy for the drive unit
Information technology and tooling	325	140	132.1	spares buy for the drive unit
Professional services	217	70	210.0	event spend deferred
Recruiting fees	434	105	313.3	overtime in the closing weeks
Travel and subsistence	217	140	55.0	headcount ramp
Insurance and compliance	325	704	-53.8	test cell deferred
Test and validation materials	434	492	-11.8	second hub fit-out
Capital expenditure	2063	352	486.1	overtime in the closing weeks

Decisions, owners and dates recorded in Q1 2026

The action list was pruned deliberately, because a list nobody believes is worse than a shorter list that gets done. Owen Castellanos, VP Customer Success, took the largest open item. One item closed with measured evidence. The cadence changed to six-weekly after this. This was the final monthly of the cycle. The oldest item dates from two sessions

back. Three items carried forward into 2026-05-14. On-time delivery 96 percent, Tessellate Controls the outlier. Harness and electrical ran 97 percent first pass. Salinas Valley carries 37 of the queued fleet. Capital spend 890 thousand this period.

Items closed this period were closed with named evidence: a report, a signed document or a measured result. Owen Castellanos, VP Customer Success, took the largest open item. One item closed with measured evidence. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Owners were named and dates set against 2026-05-14. Emil Farkas inherited an item after a role change. Capital spend 890 thousand this period. Utilisation 85 percent, Crew Bravo fully committed. Owen Castellanos owns the Lidar assemblies recovery. Crew Bravo closed Grand Forks without a second visit.

Cross-functional items were given a single owner rather than two, after a period where shared ownership meant no ownership. The oldest item dates from two sessions back. Emil Farkas inherited an item after a role change. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Owen Castellanos, VP Customer Success, took the largest open item. Three items carried forward into 2026-05-14. Harness and electrical ran 97 percent first pass. Scrap 13 thousand, Calder Battery Works contributing. Owen Castellanos owns the Lidar assemblies recovery. Utilisation 85 percent, Crew Bravo fully committed.

Evidence of closure means a document, a measurement or a signature, and status colours no longer qualify. Owners were named and dates set against 2026-05-14. Two due dates were changed in the room. The cadence changed to six-weekly after this. This was the final monthly of the cycle. Owen Castellanos, VP Customer Success, took the largest open item. Emil Farkas inherited an item after a role change. Harness and electrical ran 97 percent first pass. Build plan 112, actual 110, Lidar assemblies short. Days outstanding 39, Havering Logistics the largest balance. Uptime 98.6 percent, interventions every 433 hours.

Supplementary notes were retained in the working file rather than in the circulated packet, at the chair's request. Pinebank Distribution is named only in the working file. Figures here match the packet issued for Q1 2026. The cadence changed to six-weekly after this. This was the final monthly of the cycle. This detail was not read into the Q1 2026 session. The chair reviewed it before 2026-05-14. Queue depth 37 sites, Waukegan at the front. Depot fill 98 percent at Reno depot. Owen Castellanos owns the Lidar assemblies recovery. Yield 97 percent, scrap 13 thousand in Harness and electrical.

Working notes from the pre-meeting between the chair and the constrained function are summarised below. The chair reviewed it before 2026-05-14. It concerns Waukegan and, in part, Ashland Park. The cadence changed to six-weekly after this. This was the final monthly of the cycle. The material was retained after the 2026-04-23 meeting closed. Figures here match the packet issued for Q1 2026. Work in process 46 units, mostly Lidar assemblies. Scrap 13 thousand, Calder Battery Works contributing. Go-live cycle 37 days at Ashland Park. Capital spend 890 thousand this period.

Open and closed actions carried through Q1 2026

Reference	Item	Owner	Original due	Current due	Disposition
OR-1508	Publish runway triggers with named observables	Ingrid Solberg	2026-04-14	2026-05-05	Closed; trigger register with dated arming decisions
OR-2201	Make floor-marking survey a pre-install readiness gate	Priya Ramanathan	2026-04-20	2026-05-11	Closed; gate checklist in the deployment playbook revision
OR-2420	Publish the supplier scorecard with weighted criteria	Hollis Bramwell	2026-04-13	2026-05-04	Open; outstanding
OR-2423	Cost the second-shift ramp against overtime	Marcus Peil	2026-04-19	2026-05-10	In progress; outstanding
OR-2426	Reconcile the two uptime definitions with the customer	Owen Castellanos	2026-04-25	2026-05-16	Closed; report filed
OR-2429	Complete process capability study on the bond operation	Emil Farkas	2026-05-01	2026-05-22	Open; outstanding
OR-2432	Stage software releases across sites rather than at once	Griffin Tasse	2026-05-07	2026-05-28	In progress; outstanding

Reference	Item	Owner	Original due	Current due	Disposition
OR-2435	Rebuild the site cost model separating retrofit from greenfield	Marcus Peil	2026-05-13	2026-06-03	Closed; report filed
OR-2438	Move collections calls to the account teams	Lucia Mbeki-Hart	2026-05-19	2026-06-09	Open; outstanding
OR-2441	Draft deposit terms for multi-site commitments	Ingrid Solberg	2026-05-25	2026-06-15	In progress; outstanding
OR-2444	Size the buffer stock on long-lead commodities	Hollis Bramwell	2026-05-31	2026-06-21	Closed; report filed
OR-2447	Measure onboarding time to productivity for crews	Renata Voss	2026-06-06	2026-06-27	Open; outstanding
OR-2450	Re-cut the stage definitions in the pipeline report	Marguerite Do	2026-06-12	2026-07-03	In progress; outstanding
OR-2453	Decide disposition of the superseded sensor head stock	Marcus Peil	2026-06-18	2026-07-09	Closed; decision recorded
OR-2456	Split the reliability backlog by build vintage	Tobias Nkemdirim	2026-06-24	2026-07-15	Open; outstanding
OR-2459	Reissue the deployment playbook network section	Priya Ramanathan	2026-06-30	2026-07-21	In progress; outstanding